

Part 1: Multi-Dimensional OLAP Operations Case Analysis

Request A: Slice

This is a cutting operation. The report still covers the full fiscal year 2025, all products and customer segments, and management is only constraining one dimension — Geography — to a single member value ("North"). The slice is also different from a dice because a dice is taking more than one dimension to be reduced at a time, but a slice is taking only one.

Request B: Drill-down

This is a drill down operation. Clicking into North takes the executive to a more detailed level (monthly totals by individual Store), which can be summarized back up to the Region level by clicking into the store. This operation goes from the coarser (Region and Year) to finer (Store and Month) level of the hierarchies simultaneously (Geography goes down and Time goes down), the inverse of roll-up.

Request C: Dice

This is a dice operation. Request A has multiple dimensions being filtered, Product (Electronics and Apparel), Customer Segment (Corporate only), Geography (East and West), and Time (Q1). Restricting two or more dimensions at once (several with multiple members) creates a not-necessarily-rectangular sub-cube, rather than a flat cross section as would be created by a dice.

Request D: Roll-up

This is a roll up operation. Both the Time hierarchy (Day → Month) and the Geography hierarchy (Store → District) are automatically ascending every day. The monthly-per-store totals are created using SUM() and GROUP BY, which is the same operation that was implemented in Part

2's SQL query: the MonthlyStoreRevenue CTE takes row-level fact_sales transactions and collapses them to a monthly-per-store total, representatively implementing this roll-up operation in a relational-database. In Part 3, the MongoDB pipeline is the same: It sums up individual documents that report instances of sales for a product_category, using \$group. These demonstrate that both GROUP BY (SQL) and \$group (MongoDB) are the conventional ways that databases do a roll-up.

Request E: Pivot

This is a pivot (or rotate) operation. No data is lost and no aggregation level changes: Store Locations and Product Categories are both fully present, they just swap axis roles (rows vs. columns). This is a purely presentational change, as opposed to the other 4 operations: slice, dice, roll-up, and drill down, in that they change what data is included, or at what level of detail; pivot only changes how the same data is presented for viewing.